

IT24101148

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PS_LAB4

```
1 setwd("C:\\Users\\Sachith\\Desktop\\PS\\IT24101148")
2
3 branch_data <- read.table("Exercise.txt", header = TRUE, sep = ",")
4 head(branch_data)
5 str(branch_data)
6
7
8 boxplot(branch_data$Sales_X1,
9         main = "Boxplot for Sales",
10        ylab = "Sales",
11        col = "lightblue")
12
13
14 summary(branch_data$Sales_X1)
15
16 fivenum(branch_data$Advertising_X2)
17 IQR(branch_data$Advertising_X2)
18
19
20 find_outliers <- function(x) {
21   Q1 <- quantile(x, 0.25)
22   Q3 <- quantile(x, 0.75)
23   IQR_value <- IQR(x)
24   lower_bound <- Q1 - 1.5 * IQR_value
25   upper_bound <- Q3 + 1.5 * IQR_value
26   outliers <- x[x < lower_bound | x > upper_bound]
27   return(outliers)
28 }
29
30
31 outliers_years <- find_outliers(branch_data$Years_X3)
32 print(outliers_years)
33
34
```

```

> setwd("C:\\Users\\Sachith\\Desktop\\PS\\IT24101148")
>
> branch_data <- read.table("Exercise.txt", header = TRUE, sep = ",")
> head(branch_data)
  Branch Sales_X1 Advertising_X2 Years_X3
1      1      3.4           120        4
2      2      4.1           150        7
3      3      2.8            90        3
4      4      5.0           200       10
5      5      3.7           110        5
6      6      4.5           175        6
> str(branch_data)
'data.frame':   30 obs. of  4 variables:
 $ Branch      : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Sales_X1    : num  3.4 4.1 2.8 5 3.7 4.5 3 4.9 3.2 2.5 ...
 $ Advertising_X2: int  120 150 90 200 110 175 95 185 105 80 ...
 $ Years_X3    : int  4 7 3 10 5 6 2 9 4 1 ...
>
>
> boxplot(branch_data$Sales_X1,
+         main = "Boxplot for Sales",
+         ylab = "Sales",
+         col = "lightblue")
`

> summary(branch_data$Sales_X1)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 2.500  3.125   3.850   3.790  4.375   5.100
> fivenum(branch_data$Advertising_X2)
[1] 80.0 100.0 132.5 160.0 210.0
> IQR(branch_data$Advertising_X2)
[1] 57.5
>
>
> find_outliers <- function(x) {
+   Q1 <- quantile(x, 0.25)
+   Q3 <- quantile(x, 0.75)
+   IQR_value <- IQR(x)
+   lower_bound <- Q1 - 1.5 * IQR_value
+   upper_bound <- Q3 + 1.5 * IQR_value
+   outliers <- x[x < lower_bound | x > upper_bound]
+   return(outliers)
+ }
>
>
> outliers_years <- find_outliers(branch_data$Years_X3)
> print(outliers_years)
integer(0)

```

Boxplot for Sales

